

SQL Assignment

Total Marks - 50

Part 1: Schema Design & Creation (10 Marks)

1. **Create a new database** named **TechZoneDB**.
 2. **Create the following 3 tables** with appropriate Primary Keys, Foreign Keys, and Data Types:
 - **Products Table:** Needs columns for **ProductID**, **ProductName**, **Category** (e.g., Laptop, Phone), **Price**, and **StockQuantity**.
 - **Customers Table:** Needs columns for **CustomerID**, **FirstName**, **LastName**, **Email**, and **RegistrationDate**.
 - **Orders Table:** Needs columns for **OrderID**, **CustomerID** (Link to Customers), **ProductID** (Link to Products), **OrderDate**, and **Quantity**.
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Part 2: Data Population (10 Marks)

1. **Insert 5 Products:**
 - (1, 'MacBook Pro', 'Laptop', 1200.00, 10)
 - (2, 'iPhone 17', 'Phone', 999.00, 20)
 - (3, 'Dell XPS', 'Laptop', 1100.00, 8)
 - (4, 'Samsung S25', 'Phone', 850.00, 15)
 - (5, 'Sony Headphones', 'Audio', 200.00, 50)
 2. **Insert 4 Customers:**
 - (1, 'John', 'Doe', 'john@mail.com', '2026-01-10')
 - (2, 'Jane', 'Smith', 'jane@mail.com', '2026-02-15')
 - (3, 'Alice', 'Brown', 'alice@mail.com', '2026-03-20')
 - (4, 'Bob', 'White', 'bob@mail.com', '2026-04-05')
 3. **Insert 5 Orders (The Transactions):**
 - John Doe bought 1 MacBook Pro.
 - Jane Smith bought 2 iPhone 17.
 - Alice Brown bought 1 Sony Headphone.
 - Bob White bought 1 Dell XPS.
 - John Doe bought 1 Samsung S25 (He is a returning customer).
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Part 3: Data Manipulation & Maintenance (10 Marks)

1. **Price Hike:** The 'iPhone 17' price has increased by \$50. Write a query to update its price in the **Products** table.

2. **Inventory Adjustment:** A stock check revealed that 5 units of 'Sony Headphones' were damaged. Update the `StockQuantity` to subtract 5 from the current amount.
 3. **Order Cancellation:** The customer 'Alice Brown' cancelled her order for the Headphones. Write a query to delete that specific record from the `Orders` table.
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Part 4: Business Reporting & Analysis (10 Marks)

1. **Sales Report:** Write a query that shows the **Customer's Full Name**, **Product Name**, and **Order Date** for all orders.
 2. **Revenue Calculation:** Write a query to calculate the **Total Revenue** generated by "Laptop" sales.
 3. **Top Spenders:** Write a query to show **Customer Name** and the **Total Quantity** of items they have purchased, ordered from highest to lowest.
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Part 5: Advanced Engineering (10 Marks)

1. **Create a View:** Create a view named `HighValueProducts` that dynamically displays only products with a price greater than \$1000.
2. **Create a Stored Procedure:** Create a procedure named `RegisterCustomer` that accepts a First Name, Last Name, and Email, and adds them to the Customers table with the current date.